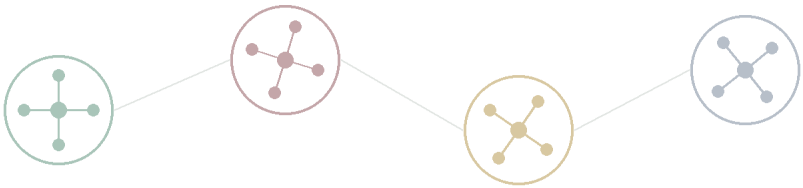
PORTFOLIO OPTIMIZATION WITH K-MEANS CLUSTERING

A Walk-Forward Study of Diversification, Portfolio Risk and Methodological Robustness

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Research question and approach. Does clustering stocks by historical return behavior improve diversification and risk-adjusted performance relative to conventional allocation methods once look-ahead bias is corrected? We study 15 U.S. equities using monthly returns from 2015–2024, compare five portfolio rules in an **82-month** walk-forward test, refit K-means only on information available at each rebalance, and evaluate CAGR, volatility, Sharpe ratio, drawdown, turnover and portfolio concentration.

1 Executive summary

Five monthly rebalanced strategies were compared: equal-weight, long-only minimum variance, long-only maximum Sharpe, inverse volatility, and K-means cluster-balanced allocation. The script requests adjusted daily closes through yfinance for 2015–2024, converts to month-end prices and monthly simple returns, and begins portfolio testing after 36 observed monthly returns. The supplied run produced 119 monthly return observations and 82 forward returns.

The reference implementation formed its K-means groups using all observed months and then reused those groups throughout its backtest. That construction exposes earlier allocations to information from later dates. The revised implementation re-estimates four clusters at every decision point using only the available history, and assigns equal capital to each cluster and equal weight within each cluster.

With stock-wise standardization preserved, K-means generated **17.50%** CAGR, 16.8% annualized volatility, **1.02** reported Sharpe, −23.6% maximum drawdown and approximately 0.20 average two-sided monthly turnover. Equal-weight delivered **20.15%** CAGR, 16.9% volatility, **1.18** Sharpe, −19.9% drawdown and approximately 0.05 turnover. Thus K-means offered similar volatility but worse realized growth, risk-adjusted return, drawdown and turnover in this sample.

The methodological lesson is: a strategy that looks compelling under one preprocessing convention can be considerably weaker when its construction is made consistent. A separate exploratory implementation standardized cross-sections rather than each stock's own time series; its 28.30% CAGR is not a valid estimate of the final original-method strategy and must not be used as the headline.

Investment interpretation

Cluster balancing is a transparent way to allocate across data-derived groups, but it is not itself a return forecast. When cluster membership changes, rebalancing can be costly. A portfolio manager should demand stability, implementation-cost analysis and repeated historical tests before treating this rule as an improvement over simple equal-weighting.

2 Data, assumptions and universe

The unchanged 15-ticker universe is AAPL, MSFT, NVDA, AMZN, GOOGL, JPM, V, JNJ, UNH, PG, KO, WMT, XOM, CAT and DIS. The script requests prices from 1 January 2015 to 31 December 2024, using auto-adjusted closes. Month-end prices are formed with resampling; percentage changes yield the monthly return matrix. In the verified research run, daily prices covered 2 January 2015 to 30 December 2024, with 119 monthly returns.

This deliberately chosen set represents multiple sectors, but it is not a randomly sampled or point-in-time investable universe. Retrospective selection of surviving, well-known firms creates selection and survivorship concerns. Results should not be generalized to a broad equity index.

Assumption	Implementation
Assets	15 fixed U.S. stocks
Return frequency	Monthly simple returns
Risk-free rate	2% annual, simplified
Formation history	36 months initially; expanding thereafter
Rebalancing	Monthly; long-only, fully invested

K-means	K=4, n_init=10, random_state=0
Costs	Not deducted from portfolio returns
Data contingency	Script has synthetic fallback if yfinance raises an exception

Reproducibility caveat: the script can silently switch to synthetic prices after a download exception; a production research workflow should fail loudly and record the price source, observation count, data snapshot and run date. The reported run is interpreted as market-data based using the previously checked 2,515 daily rows × 15 assets; the report does not assert an independently archived vendor snapshot.

3 Financial concepts and formulas

The original report begins with Markowitz portfolio theory and builds from returns, risk, covariance and correlation to portfolio construction. The same sequence is essential here: K-means is not a substitute for these fundamentals but an additional rule for grouping stocks before setting weights.

Concept	Formula	Meaning / implementation
Monthly simple return	$r_{i,t} = P_{i,t} / P_{i,t-1} - 1$	Adjusted month-end prices
Annualized return and risk	$\mu_{i,ann} = 12\,\bar{r}_i$; $\sigma_{i,ann} = \sqrt{12} \times \mathrm{std}(r_{i,t})$	Historical estimates; volatility scaling is approximate
CAGR	$(E_n/E_0)^{(12/n)} - 1$	Realized compound growth; differs from script ann.ret
Variance and covariance	$\mathrm{Var}(r_i) = E[(r_i-\mu_i)^2]$; $\Sigma_{ann} = 12\,\mathrm{Cov}(r)$	Risk and co-movement inputs
Correlation	$\rho_{ij} = \mathrm{Cov}(r_i,r_j)/(\sigma_i\sigma_j)$	Linear co-movement from −1 to +1
Two-asset variance	$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$	Diversification; XOM–JNJ example
Two-asset minimum risk	$w_1^* = (\sigma_2^2-\rho\sigma_1\sigma_2)/(\sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2)$	Unconstrained analytical weight
Markowitz	$\mu_p = w^T\mu$; $\sigma_p^2 = w^T\Sigma w$	Minimize variance; weights sum to 1 and are nonnegative
Maximum Sharpe	$(w^T\mu - rf)/\sqrt{(w^T\Sigma w)}$	rf = 2% annually
Reported Sharpe	$[((1+\bar{r}_p)^{12}-1)-0.02]/[\mathrm{std}(r_p)\sqrt{12}]$	Script-specific, nonstandard convention
Maximum drawdown	$\min_t [E_t / \max_{s\le t} E_s - 1]$	Worst historical peak-to-trough decline
Stock-wise standardization	$z_{i,t} = (r_{i,t} - \bar{r}_i)/s_i$	Fit separately for each stock using available history
K-means objective	$\sum_j \sum_{i\in C_j} \lVert z_i - m_j \rVert^2$	K=4; n_init=10; random_state=0
Cluster-balanced weight	$w_i = 1/(K\,n_c)$	25% per cluster, equally split within cluster
Concentration	$HHI = \sum_i w_i^2$; $N_{eff} = 1/HHI$	Effective holdings; mean reciprocal ≠ reciprocal of mean
Turnover	$\sum_i \lvert w_{i,new} - w_{i,drifted} \rvert$	Two-sided monthly weight changes; pre-cost

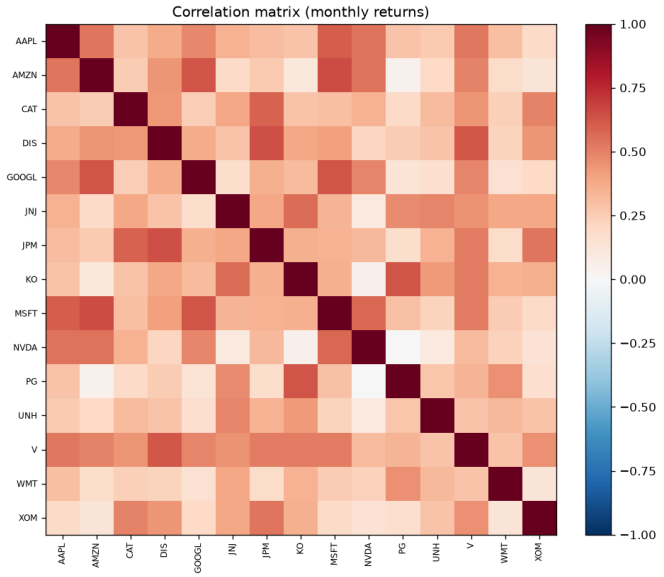


Figure 1. Full-sample monthly-return correlation matrix; descriptive rather than a live trading input.

Correlations are mostly positive but vary by stock; they describe historical co-movement, not causation or crisis protection.

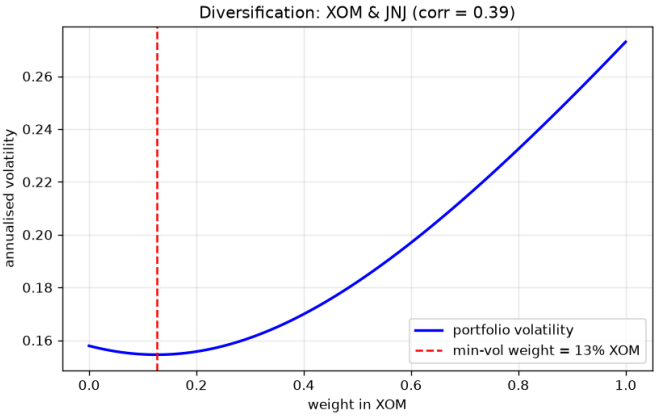


Figure 2. XOM–JNJ two-asset illustration: correlation approximately 0.39 and minimum volatility near 13% XOM.

The curve illustrates how imperfect correlation reduces volatility; it does not establish K-means outperformance.

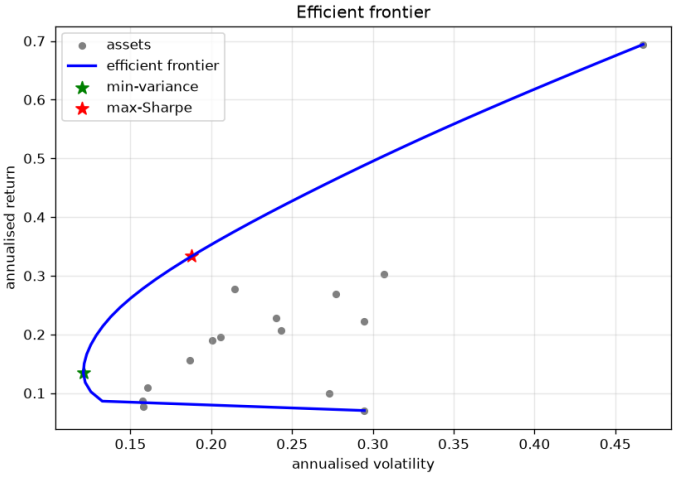
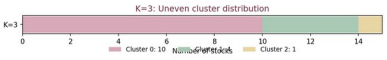
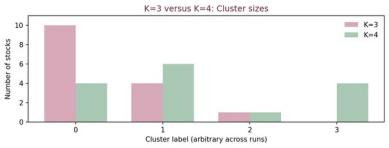
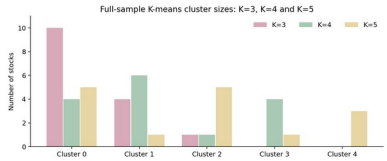


Figure 3. Mean–variance opportunity set. Only the upper branch above the global minimum-variance point is efficient.

Green marks minimum variance; red marks maximum Sharpe. The lower blue branch is inefficient.







Exploratory cluster structure | Exploratory cluster sizes | Cross-K size comparison

For cluster c containing n_c stocks, each member receives $w_i = 1/(K\,n_c)$. Each cluster therefore receives 1/K = 25% of capital, even if the clusters have different numbers of stocks. This prevents large groups from automatically dominating the portfolio, but can overweight small clusters and increase turnover as memberships change.

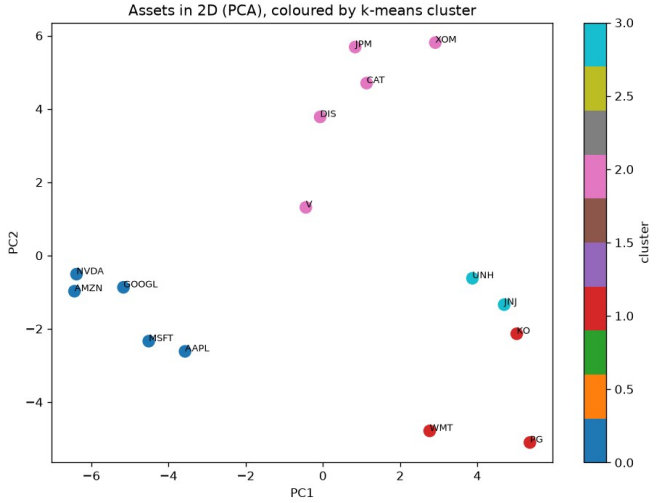


Figure 5. PCA projection of full-sample stock-wise standardized returns, colored by K-means group. Descriptive only.

The full-sample four-cluster grouping places AAPL, AMZN, GOOGL, MSFT and NVDA together; KO, PG and WMT together; CAT, DIS, JPM, V and XOM together; and JNJ and UNH together. PCA compresses 119 return dimensions into two display coordinates; distances in the plot are an approximation, not a complete representation of cluster structure.

For the actual historical-only K-means strategy, the mean largest stock weight was **13.52%**; average Herfindahl concentration HHI was **0.0793**; and the average effective number of stocks, computed as the mean of 1/HHI across 82 rebalances, was **12.72**. An equal-weight **15-stock** portfolio has HHI 0.0667 and effective holdings 15. NVDA never had a 25% weight under the corrected preprocessing.

Thus K-means was less concentrated than the earlier cross-sectionally standardized experiment, yet somewhat more concentrated than equal-weight. These are concentration measures, not evidence of lower downside risk.

4 Portfolio construction and look-ahead control

Equal-weight allocates 1/15 to each stock. Minimum variance minimizes $w^T\Sigma w$ subject to nonnegative weights summing to one. Maximum Sharpe optimizes $(w^T\mu - rf)/\sqrt{(w^T\Sigma w)}$ under the same constraints. Inverse volatility allocates in proportion to $1/\sigma_i$. Both optimization strategies estimate inputs from historical observations at each rebalance.

K-means represents each stock as a row of its historical monthly return series. Within each stock, returns are standardized by subtracting that stock's historical mean and dividing by its historical standard deviation. K-means partitions these standardized histories into four groups. Each group receives 25% of capital; a group containing m stocks gives each member 25%/m.

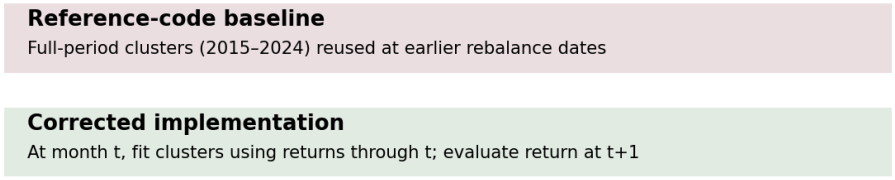
At rebalance t, the corrected function receives hist = rets.iloc[:t+1], estimates all transformations and cluster

labels from hist, constructs weights w_t, and applies them to the next month's return rets.iloc[t+1]. It repeats the fit at every step. The descriptive full-period clustering and PCA plot remain outside this trading loop and are not used to choose historical weights.

Why the distinction matters

A5. Look-ahead bias: full-sample versus historical-only clusters

Methodological diagram retained from Wednesday. The final corrected backtest uses historical-only K-means and original stock-wise standardization.



Historical figure A5. Look-ahead bias: full-sample versus historical-only clusters.

The reference script used cluster labels estimated from the entire 2015–2024 sample inside an otherwise walk-forward backtest. A portfolio in 2018 cannot know how a stock will behave in 2023. Using those later observations in cluster assignment creates look-ahead bias even if next-month returns are applied after weight selection. The corrected implementation addresses that specific leakage, but not all forms of research bias.

Turnover definition

Before each new rebalance, previous target holdings are drifted by the preceding realized monthly asset returns. The script measures the sum of absolute differences between new target and drifted weights. This is a two-sided turnover measure; a one-way traded fraction would be half that number under standard assumptions. No commission, spread, slippage or tax is charged.

5 Backtest results

Strategy	CAGR	Vol.	Sharpe*	Max DD	Turnover†
Equal-weight	**20.15%**	**16.9%**	1.18	**−19.9%**	0.05
Minimum variance	**13.66%**	**14.5%**	0.88	**−16.6%**	0.10
Maximum Sharpe	**24.89%**	**19.0%**	1.32	**−26.6%**	0.09
Inverse volatility	**17.56%**	**15.5%**	1.09	**−17.6%**	0.04
K-means (K=4)	**17.50%**	**16.8%**	1.02	**−23.6%**	0.20

* The script computes its Sharpe ratio as [((1 + average monthly return)^12 − 1) − 2%] / (monthly return standard deviation × √12). This is a nonstandard annualized-return convention, used consistently across all five strategies. † Average two-sided monthly turnover, approximately rounded. All figures are pre-cost.

CAGR is calculated from the start and end of each portfolio equity curve: (ending wealth / beginning wealth)^(12 / number of tested months) − 1. Unlike the script's "ann.ret" field, this is a realized compound growth measure. The backtest covers 82 next-month returns; CAGR values were recomputed from the final corrected equity curves.

Maximum Sharpe had the highest CAGR and reported Sharpe, but also the deepest drawdown among these five strategies. Minimum variance had the lowest drawdown and the lowest CAGR. These differences emphasize that portfolio selection depends on an investor's objective and loss tolerance, not only return ranking.

K-means underperformed equal-weight by approximately 2.65 percentage points of CAGR and 0.16 Sharpe units. Its turnover was about four times as large. Similar volatility did not translate into better risk-adjusted returns or drawdown protection.

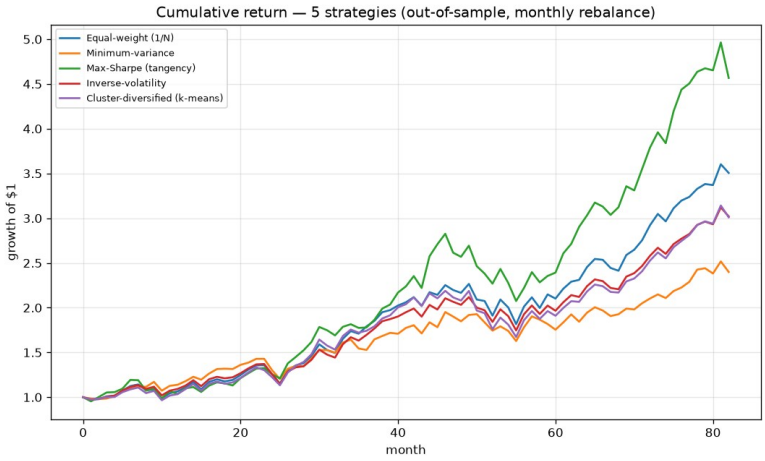


Figure 4. Growth of $1 over 82 walk-forward months, before trading costs.

Maximum Sharpe finishes highest; K-means trails equal-weight. These are historical, pre-cost outcomes, not forecasts.

6 Robustness, limitations and next experiments

Exploratory K=3, K=4 and K=5 cluster charts are descriptive only: they use different preprocessing and cannot establish a performance ranking. Cluster sizes and singleton groups highlight concentration sensitivity.

Preprocessing matters: cross-sectional standardization yielded 28.30% CAGR versus 17.50% under stock-wise standardization. Because this changes the stock representation and weights, the gap cannot be attributed solely to correcting look-ahead bias.

Limitations include ex-post stock selection, regime shifts, noisy optimization inputs, fixed K=4, no transaction costs, and uncertain price-data provenance if fallback data are used. Findings describe this 15-stock historical sample, not future returns.

Recommended next tests

Next: test K=3 and K=5 using identical historical-only preprocessing; estimate trading costs and cluster stability; include a point-in-time universe, uncertainty estimates and a separate test period. These are proposed, not completed, analyses.

7 Interpretation of the original and corrected evidence

7.1 The original report's investment claim

The original October 2026 project report describes the cluster-diversified strategy as offering an attractive all-round combination of a roughly 1.15 Sharpe ratio, −16.6% maximum drawdown and low turnover near 0.04. Those figures belong to the original implementation and are not results of the historically refitted strategy. The original narrative also calls the backtest free of look-ahead bias. However, the original code forms cluster labels from the complete return matrix before the walk-forward loop and then reuses those labels when calculating earlier-period portfolio weights. That discrepancy between the stated methodology and implementation is the principal audit finding.

7.2 Isolating the methodological changes

There are three distinct objects that must not be merged: (a) the original full-sample-cluster implementation,

(b) an exploratory historical-only implementation with month-wise cross-sectional standardization, and (c) the final historical-only implementation with the original stock-wise standardization. The exploratory version produced 28.30% CAGR and unusually concentrated allocations; its inputs were standardized across stocks for each month. The final version instead standardized each stock across its own historical months, producing **17.50%** CAGR and substantially lower concentration. Because both the timing and the representation differ across some comparisons, the change in return cannot be attributed solely to removal of look-ahead bias.

The final implementation provides the cleanest test of the original portfolio rule with historical-only cluster formation, because it holds the original stock universe, number of clusters, random seed and cluster-balancing rule fixed. It is therefore appropriate to compare its **17.50%** CAGR with the four conventional strategies under the same corrected script. It is not appropriate to claim the experiment proves that one particular preprocessing choice will always dominate another or that K-means is generally inferior to traditional methods.

7.3 What a portfolio manager would ask

A portfolio manager would test whether the strategy adds value after transaction costs, whether clusters remain stable, and whether results hold across market regimes, alternative stock universes, cluster counts and independent test periods. Comparisons with broad indexes, equal-risk and sector-neutral benchmarks would help separate genuine diversification benefits from stock-specific return exposure.

With a 13.52% mean largest position, HHI of 0.0793 and 12.72 effective holdings, K-means avoids the exploratory singleton concentration but remains more concentrated than equal-weight. Drawdown, sector exposure and turnover also matter.

7.4 Research integrity and presentation

An academically defensible report separates the original author's stated claim, what the supplied code actually computes, the author's own methodological modification and the outcomes observed after that modification. It retains negative findings, labels descriptive full-sample illustrations, reports the metric definitions exactly and avoids invented sensitivity results. This study does not claim a corrected K=3 or K=5 backtest, a statistical significance test, a cost-adjusted return or a truly prospective portfolio track record. Those remain open tasks.

Source and reproducibility notes

Primary research basis: Abhay Dodiya, "Portfolio Optimisation with K-Means Clustering," October 2026, and supplied Python implementation; revised implementation: run_project_corrected.py (uploaded 23 October 2026). Data requested from Yahoo Finance using yfinance. Five PNG figures were generated by the revised script. Report calculations and concentration summaries were checked in Jupyter from the final revised backtest.

Conclusion

This study asked whether K-means clustering improves portfolio diversification and risk-adjusted performance once look-ahead bias is addressed. Using the original 15-stock universe and stock-wise standardization, the historical-only K-means strategy achieved a 17.50% realized CAGR and 1.02 reported Sharpe ratio, compared with 20.15% and 1.18 for equal-weight. Its average effective number of holdings was 12.72, but its higher turnover and weaker drawdown outcome limited its practical appeal. The central contribution is therefore methodological: correcting information timing and keeping preprocessing consistent materially changes the investment conclusion. K-means remains a useful grouping tool, but these results do not establish it as a superior allocation strategy; broader samples, alternative cluster counts and transaction-cost tests are needed.